Choosing the resolution moves the answer: a measured inference bias in SPH impact modelling

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Abstract

Smoothed-particle impact simulations are routinely run at the coarsest resolution that fits the budget, and the resulting discretisation error is folded into the likelihood as extra variance. We show, on a 96-point production campaign for the DART impact on Dimorphos, that this is the wrong operation. Re-running nine of the design points at half the particle spacing moves the momentum enhancement in the *same direction at all nine points* (median ratio 1.139, sign test $p = 0.002$): the discretisation error is a bias, not scatter. Carrying it as variance leaves the posterior centred on the coarse prediction, so the interior parameters absorb the shift. Correcting it instead with a Kennedy–O'Hagan autoregressive surrogate moves the inferred matrix strength from 18.7 Pa to 159.7 Pa, a 1.88σ shift, and the shift lands on the one direction the single observable can constrain at all. Neither value is converged; the point is that the choice of resolution, not the data, selects between them.

1 The problem

A kinetic impact leaves one well-measured scalar, the momentum enhancement β. Inferring interior properties from it requires a forward model, and that model is discretised. The usual treatment adds a resolution term σ_{res} to the error budget.

That treatment assumes the discretisation error behaves like noise. If it is instead a systematic offset, widening the likelihood does not move the posterior's centre: the parameters must absorb the offset. One then believes a wrong interior precisely because it reproduces the right β.

2 Measurement

The production campaign is 96 Latin-hypercube points over boulder porosity α_b, matrix strength Y_0 and boulder volume fraction f, run with a GPU SPH solver (FP64, Wendland C2, Tillotson, P–α porosity, Grady–Kipp damage, self-gravity) on a triaxial $88.5 \times 87.0 \times 58.0$ m body at 7.0 m base spacing. Nine points of a maximin subset, fixed before any data existed, were re-run at 3.5 m: 441 GPU-hours.

Table 1 gives the result: 9 of 9 in the same direction, median ratio 1.139. A one-sided sign test gives $p = 2^{-9} = 0.002$. The coarse campaign underestimates β by about 14%, systematically.

Table 1. Fine against coarse at identical θ. Every point moves the same way.

point	coarse	fine	ratio
11	1.583	1.743	1.101
20	5.040	5.743	1.139
33	3.879	4.841	1.248
44	1.728	1.783	1.032
45	2.644	2.976	1.126
57	3.524	4.671	1.326
58	2.758	3.241	1.175
62	2.558	2.731	1.068
74	5.016	5.775	1.151

Table 2. Posterior mode, uncorrected against corrected. Shift in units of the uncorrected posterior's own standard deviation.

axis	uncorrected	corrected	shift
α_b	1.269	1.036	2.74σ
Y_0 [Pa]	18.7	159.7	1.88σ
f	0.05	0.05	0.00σ

3 Correction

Because the error is systematic it can be modelled. We fit a Kennedy–O'Hagan autoregressive surrogate, $y_{\mathrm{fine}}(x) = \rho\, f_{\mathrm{coarse}}(x) + \delta(x)$, on the nine nested points, obtaining $\rho = 0.820$ for β and $\rho = 1.180$ for log ejecta mass, both interior to the search grid.

Leave-one-out across the nine points, against the fine value: RMS error falls from 0.656 uncorrected to 0.262 corrected, a factor 2.51, and the corrected surrogate wins at all nine points individually.

4 Consequence for the inference

Both posteriors were then computed in a single run sharing the design, the observable transform, the residual correlation, the error model, the grid and the likelihood. Only the surrogate mean differs, so any difference cannot be attributed elsewhere.

The inferred matrix strength moves by a factor of 8.6 (Table 2). The shift falls on Y_0, which is the only direction this observable constrains: a Fisher decomposition over the same 96 points gives posterior widths of 0.142, 0.930 and 1.000 of the prior along the three principal directions, with the learnable one aligned with Y_0 at 90 of 96 points. A shift along an unconstrained axis

would not matter. This one does.

5 What this does not show

The fine runs are not converged either: differences grow with refinement rather than settling, so 159.7 Pa is what one further step of refinement says, not where the sequence lands. We therefore report both numbers and treat neither as the interior of Dimorphos. The modes for α_b and f sit against their prior bounds and are not estimates. The nested design was planned at twelve points and nine were run, so $\delta(x)$ is extrapolated near the design edges.

The claim is narrower than a value and, we think, more useful: in this class of problem the resolution decision is not a nuisance absorbed by the error bars. It selects the answer, and it should be reported as part of the result.

6 Availability

Code, the locked decision rules, the measured outputs and the scripts that regenerate every number above are public at github.com/Yagiz-D-Ekrem-Dolar/dart-rift. The gate used to decide whether the corrected posterior may be promoted was written in a committed decision record before the fine data existed; the records behind this note are `KAYIT-099` and `KAYIT-100`.